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OIL EXPLORATION

Introduction

Crude oils and natural gases are mixtures of hydrocarbon molecules (organic compounds of carbon and hydrogen atoms) containing from 1 to 60 carbon atoms. The properties of these hydrocarbons depend on the number and arrangement of the carbon and hydrogen atoms in their molecules. The basic hydrocarbon molecule is 1 carbon atom linked with 4 hydrogen atoms (methane). All other variations of petroleum hydrocarbons evolve from this molecule. Hydrocarbons containing up to 4 carbon atoms are usually gases; those with 5 to 19 carbon atoms are usually liquids; and those with 20 or more are solids. In addition to hydrocarbons, crude oils and natural gases contain sulphur, nitrogen and oxygen compounds together with trace quantities of metals and other elements.

Crude oil and natural gas are believed to have been formed over millions of years by the decay of vegetation and marine organisms, compressed under the weight of sedimentation. Because oil and gas are lighter than water, they rose up to fill the voids in these overlying formations. This upward movement stopped when the oil and gas reached dense, overlying, impervious strata or nonporous rock. The oil and gas filled the spaces in porous rock seams and natural underground reservoirs, such as saturated sands, with the lighter gas on top of the heavier oil. These spaces were originally horizontal, but shifting of the earth's crust created pockets, called faults, anticlines, salt domes and stratigraphic traps, where the oil and gas collected in reservoirs.

Shale Oil

Shale oil, or kerogen, is a mixture of solid hydrocarbons and other organic compounds containing nitrogen, oxygen and sulphur. It is extracted, by heating, from a rock called oil shale, yielding from 15 to 50 gallons of oil per ton of rock.

Exploration and production is the common terminology applied to that portion of the petroleum industry which is responsible for exploring for and discovering new crude oil and gas fields, drilling wells and bringing the products to the surface. Historically, crude oil, which had naturally seeped to the surface, was collected for use as medicine, protective coatings and fuel for lamps. Natural gas seepage was recorded as fires burning on the surface of the earth. It was not until 1859 that methods of drilling and obtaining large commercial quantities of crude oil were developed.

Crude oil and natural gas are found throughout the world, beneath both land and water, as follows:

1. Western Hemisphere Intercontinental Basin (US Gulf Coast, Mexico, Venezuela)
2. Middle East (Arabian Peninsula, Persian Gulf, Black and Caspian Seas)
3. Indonesia and South China Sea
4. North and West Africa (Sahara and Nigeria)
5. North America (Alaska, Newfoundland, California and Mid-continent United States and Canada)
6. Far East (Siberia and China)
7. North Sea.

Classification of crude oils

Crude oils are complex mixtures containing many different, individual hydrocarbon compounds; they differ in appearance and composition from one oil field to another, and sometimes are even different from wells relatively near one another. Crude oils range in consistency from watery to tar-like solids, and in colour from clear to black. An “average” crude oil contains about 84% carbon; 14% hydrogen; 1 to 3% sulphur; and less than 1% of nitrogen, oxygen, metals and salts.

Crude source and name - Nigerian Light, Saudi Light, Saudi Heavy, Venezuela Heavy, Venezuela Light, USA Midcontinental Sweet, USA West Texas Sour, North Sea Brent.

Relatively simple crude-oil assays are used to classify crude oils as paraffinic, naphthenic, aromatic or mixed, based on the predominant proportion of similar hydrocarbon molecules. Mixed-base crudes have varying amounts of each type of hydrocarbon. One assay method (US Bureau of Mines) is based on distillation, and another method (UOP "K" factor) is based on gravity and boiling points. More comprehensive crude assays are conducted to determine the value of the crude (i.e., its yield and quality of useful products) and processing parameters. Crude oils are usually grouped according to yield structure, with high-octane gasoline being one of the more desirable products. Refinery crude oil feed stocks usually consist of mixtures of two or more different crude oils.

Crude oils are also defined in terms of API (specific) gravity. For example, heavier crude oils have low API gravities (and high specific gravities). A low-API gravity crude oil may have either a high or low flashpoint, depending on its lightest ends (more volatile constituents). Because of the importance of temperature and pressure in the refining process, crude oils are further classified as to viscosity, pour points and boiling ranges. Other physical and chemical characteristics, such as colour and carbon residue content, are also considered. Crude oils with high carbon, low hydrogen and low API gravity are usually rich in aromatics; while those with low carbon, high hydrogen and high API gravity are usually rich in paraffins.

Crude oils which contain appreciable quantities of hydrogen sulphide or other reactive sulphur compounds are called “sour.” Those with less sulphur are called “sweet.” Some exceptions to this rule are West Texas crudes (which are always considered “sour” regardless of their H_2S content) and Arabian high-sulphur crudes (which are not considered “sour” because their sulphur compounds are not highly reactive).

Composition of crude oil and natural gas:

HYDROCARBONS

Paraffins: The paraffinic saturated chain type hydrocarbon (aliphatic) molecules in crude oil have the formula C_nH_{2n+2}, and can be either straight chains (normal) or branched chains (isomers) of carbon atoms. The lighter, straight chain paraffin

molecules are found in gases and paraffin waxes. The branched chain paraffins are usually found in heavier fractions of crude oil and have higher octane numbers than normal paraffins.

Aromatics: Aromatics are unsaturated ring type hydrocarbon (cyclic) compounds. Naphthalenes are fused double ring aromatic compounds. The most complex aromatics, polynuclears (three or more fused aromatic rings), are found in heavier fractions of crude oil.

Naphthenes: Naphthenes are saturated ring type hydrocarbon groupings, with the formula C_nH_{2n}, arranged in the form of closed rings (cyclic), found in all fractions of crude oil except the very lightest. Single ring naphthenes (mono-cycloparaffins) with 5 and 6 carbon atoms predominate, with two ring naphthenes (dicycloparaffins) found in the heavier ends of naphtha.

NON-HYDROCARBONS

Sulphur and sulphur compounds: Sulphur is present in natural gas and crude oil as hydrogen sulphide (H_2S), as compounds (thiols, mercaptans, sulphides, polysulphides, etc.) or as elemental sulphur. Each gas and crude oil has different amounts and types of sulphur compounds, but as a rule the proportion, stability and complexity of the compounds are greater in heavier crude oil fractions.

Sulphur compounds called mercaptans, which exhibit distinct odours detectable at very low concentrations, are found in gas, petroleum crude oils and distillates. The most common are methyl and ethyl mercaptans. Mercaptans are often added to commercial gas (LNG and LPG) to provide an odour for leak detection.

The potential for exposure to toxic levels of H_2S exists when working in drilling, production, transportation and processing crude oil and natural gas. The combustion of petroleum hydrocarbons containing sulphur produces undesirables such as sulphuric acid and sulphur dioxide.

Oxygen compounds: Oxygen compounds, such as phenols, ketones and carboxylic acids, are found in crude oils in varying amounts.

Nitrogen compounds: Nitrogen is found in lighter fractions of crude oil as basic compounds, and more often in heavier fractions of crude oil as non-basic compounds which may also include trace metals.

Trace metals: Trace amounts, or small quantities of metals, including copper, nickel, iron, arsenic and vanadium, are often found in crude oils in small quantities.

Inorganic salts: Crude oils often contain inorganic salts, such as sodium chloride, magnesium chloride and calcium chloride, suspended in the crude or dissolved in entrained water (brine).

Carbon dioxide: Carbon dioxide may result from the decomposition of bicarbonates present in, or added to crude, or from steam used in the distillation process.

Naphthenic acids: Some crude oils contain naphthenic (organic) acids, which may become corrosive at temperatures above 232 °C when the acid value of the crude is above a certain level.

Normally occurring radioactive materials: Normally occurring radioactive materials (NORMs) are often present in crude oil, in the drilling deposits and in the drilling mud, and can present a hazard from low levels of radioactivity.

Compressed Natural Gas and Liquefied Hydrocarbon Gases:

The composition of naturally occurring hydrocarbon gases is similar to crude oils in that they contain a mixture of different hydrocarbon molecules depending on their source. They can be extracted as natural gas (almost free of liquids) from gas fields; petroleum-associated gas which is extracted with oil from gas and oil fields; and gas from gas condensate fields, where some of the liquid components of oil convert into the gaseous state when pressure is high (10 to 70 mPa).

When the pressure is decreased (to 4 to 8 mPa) condensate containing heavier hydrocarbons separates from the gas by condensation. Gas is extracted from wells reaching up to 4 miles (6.4 km) or more in depth, with seam pressures varying from 3 mPa up to as high as 70 mPa.

Natural gas contains 90 to 99% hydrocarbons, which consist predominately of methane (the simplest hydrocarbon) together with smaller amounts of ethane, propane and butane. Natural gas also contains traces of nitrogen, water vapour, carbon dioxide, hydrogen sulphide and occasional inert gases such as argon or helium. Natural gases containing more than 50 g/m^3 of hydrocarbons with molecules of three or more carbon atoms (C_3 or higher) are classified as "lean" gases.

Depending how it is used as a fuel, natural gas is either compressed or liquefied. Natural gas from gas and gas condensate fields is processed in the field to meet specific transportation criteria before being compressed and fed into gas pipelines.

This preparation includes removal of water with driers (dehydrators, separators and heaters), oil removal using coalescing filters, and the removal of solids by filtration. Hydrogen sulphide and carbon dioxide are also removed from natural gas, so that they do not corrode pipelines and transportation and compression equipment. Propane, butane and pentane, present in natural gas, are also removed before transmission so they will not condense and form liquids in the system. (See the section "Natural gas production and processing operations.")

Natural gas is transported by pipeline from gas fields to liquefication plants, where it is compressed and cooled to approximately –162 °C to produce liquefied natural gas (LNG). The composition of LNG is different from natural gas due to the removal of some impurities and components during the liquefaction process.

LNG is primarily used to augment natural gas supplies during peak demand periods and to supply gas in remote areas away from major pipelines. It is regasified by adding nitrogen and air to make it comparable to natural gas before being fed into gas supply lines. LNG is also used as a motor-vehicle fuel as an alternative to gasoline.

Petroleum-associated gases and condensate gases are classified as "rich" gases, because they contain significant amounts of ethane, propane, butane and other saturated hydrocarbons. Petroleum-associated and condensate gases are separated and liquefied to produce liquefied petroleum gas (LPG) by compression, adsorption, absorption and

cooling at oil and gas process plants. These gas plants also produce natural gasoline and other hydrocarbon fractions.

Unlike natural gas, petroleum-associated gas and condensate gas, oil processing gases (produced as by-products of refinery processing) contain considerable amounts of hydrogen and unsaturated hydrocarbons (ethylene, propylene and so on). The composition of oil processing gases depends upon each specific process and the crude oils used. For example, gases obtained as a result of thermal cracking usually contain significant amounts of olefins, while those obtained from catalytic cracking contain more isobutanes. Pyrolysis gases contain ethylene and hydrogen.

Combustible natural gas, with a calorific value of 35.7 to 41.9 MJ/m^3 (8,500 to 10,000 kcal/m^3), is primarily used as a fuel to produce heat in domestic, agricultural, commercial and industrial applications. The natural gas hydrocarbon also is used as feedstock for petrochemical and chemical processes. Synthesis gas (CO + H_2) is processed from methane by oxygenation or water vapour conversion, and used to produce ammonia, alcohol and other organic chemicals.

Compressed natural gas (CNG) and liquefied natural gas (LNG) are both used as fuel for internal combustion engines. Oil processing liquefied petroleum gases (LPG) have higher calorific values of 93.7 MJ/m^3 (propane) (22,400 kcal/m^3) and 122.9 MJ/m^3 (butane) (29,900 kcal/m^3) and are used as fuel in homes, businesses and industry as well as in motor vehicles (NFPA 1991). The unsaturated hydrocarbons (ethylene, propylene and so on) derived from oil processing gases may be converted into high-octane gasoline or used as raw materials in the petrochemical and chemical-processing industries.

Properties of Hydrocarbon Gases:

According to the US National Fire Protection Association, flammable (combustible) gases are those which burn in the concentrations of oxygen normally present in air. The burning of flammable gases is similar to that of flammable hydrocarbon liquid vapours, as a specific ignition temperature is needed to initiate the burning reaction and each will burn only within a certain defined range of gas-air mixtures. Flammable liquids

have a flashpoint (the temperature (always below the boiling point) at which they emit sufficient vapours for combustion). There is no apparent flashpoint for flammable gases, as they are normally at temperatures above their boiling points, even when liquefied, and are therefore always at temperatures well in excess of their flashpoints.

The US National Fire Protection Association (1976) defines compressed and liquefied gases, as follows:

"Compressed gases are those which at all normal atmospheric temperatures inside their containers, exist solely in the gaseous state under pressure."

"Liquefied gases are those which at normal atmospheric temperatures inside their containers, exist partly in the liquid state and partly in the gaseous state, and are under pressure as long as any liquid remains in the container."

The major factor which determines the pressure inside the vessel is the temperature of the liquid stored. When exposed to the atmosphere, the liquefied gas very rapidly vaporizes, travelling along the ground or water surface unless dispersed into the air by wind or mechanical air movement. At normal atmospheric temperatures, about one-third of the liquid in the container will vaporize.

Flammable gases are further classified as fuel gas and industrial gas. Fuel gases, including natural gas and liquefied petroleum gases (propane and butane), are burned with air to produce heat in ovens, furnaces, water heaters and boilers. Flammable industrial gases, such as acetylene, are used in processing, welding, cutting and heat-treating operations. The differences in properties of liquefied natural gas (LNG) and liquefied petroleum gases (LPG).

Searching for Oil and Gas:

The search for oil and gas requires a knowledge of geography, geology and geophysics. Crude oil is usually found in certain types of geological structures, such as anticlines, fault traps and salt domes, which lie under various terrains and in a wide range of climates. After selecting an area of interest, many different types of geophysical surveys

are conducted and measurements performed in order to obtain a precise evaluation of the subsurface formations, including:

Magnetometric surveys: Magnetometers hung from airplanes measure variations in the earth's magnetic field in order to locate sedimentary rock formations which generally have low magnetic properties when compared to other rocks.

Aerial photogrammetric surveys: Photographs taken with special cameras in airplanes, provide three-dimensional views of the earth which are used to determine land formations with potential oil and gas deposits.

Gravimetric surveys: Because large masses of dense rock increase the pull of gravity, gravimeters are used to provide information regarding underlying formations by measuring minute differences in gravity.

Seismic surveys: Seismic studies provide information on the general characteristics of the subsurface structure. Measurements are obtained from shock waves generated by setting off explosive charges in small-diameter holes, from the use of vibrating or percussion devices on both land and in water, and from underwater blasts of compressed air. The elapsed time between the beginning of the shock wave and the return of the echo is used to determine the depth of the reflecting substrata. The recent use of super-computers to generate three-dimensional images greatly improves evaluation of seismic test results.

Radiographic surveys: Radiography is the use of radio waves to provide information similar to that obtained from seismic surveys.

Stratigraphic surveys: Stratigraphic sampling is the analysis of cores of subsurface rock strata for traces of gas and oil. A cylindrical length of rock, called a core, is cut by a hollow bit and pushed up into a tube (core barrel) attached to the bit. The core barrel is brought to the surface and the core is removed for analysis.

When the surveys and measurements indicate the presence of formations or strata which may contain petroleum, exploratory wells are drilled to determine whether or

not oil or gas is actually present and, if so, whether it is available and obtainable in commercially viable quantities.

Offshore Operations

Although the first offshore oil well was drilled in the early 1900s off of the coast of California, the beginning of modern marine drilling was in 1938, with a discovery in the Gulf of Mexico, 1 mile (1.6 km) from the US coastline. After the Second World War, offshore drilling expanded quickly, first in shallow waters adjacent to known land-based production areas, and then to other shallow and deep water areas around the world, and in climates varying from the Arctic to the Persian Gulf. In the beginning, offshore drilling was possible only in water depths of about 91 m; however, modern platforms are now able drill in waters over 3.2 km deep. Offshore oil activities include exploration, drilling, production, processing, underwater construction, maintenance and repair, and the transport of the oil and gas to shore by ship or pipeline.

Offshore platforms

Drilling platforms support drilling rigs, supplies and equipment for offshore or inland water operations, and range from floating or submergible barges and ships, to fixed-in-place platforms on steel legs used in shallow waters, to large, buoyant, reinforced concrete, gravity-type platforms used in deep waters. After the drilling is completed, marine platforms are used to support production equipment. The very largest production platforms have accommodations for over 250 crew members and other support personnel, heliports, processing plants and crude oil and gas condensate storage capability.

Typically, with deep water floating platform drilling, the wellhead equipment is lowered to the ocean floor and sealed to the well casing. The use of fibre-optic technology allows a large, central platform to remotely control and operate smaller satellite platforms and sub-sea templates. Production facilities on the large platform process the crude oil, gas and condensate from the satellite facilities, before it is shipped on-shore.

Crude oil and gas production and conservation

Producing oil is basically a matter of displacement by either water or gas. At the time of initial drilling, almost all crude oil is under pressure. This natural pressure decreases as oil and gas is removed from the reservoir, during the three phases of a reservoir's life.

1. During the first phase, flush production, the flow is governed by the natural pressure in the reservoir which comes from dissolved gas in the oil, gas trapped under pressure above the oil and hydraulic pressure from water trapped under the oil.
2. Artificial lift, the second phase, involves pumping pressurized gas into the reservoir when the natural pressure is expended.
3. Phase three, stripper or marginal production, occurs when wells only produce intermittently.

Originally there was little understanding of the forces which affected oil and gas production. The study of oil and gas reservoir behaviour began at the beginning of the 20th century, when it was discovered that pumping water into a reservoir increased production. At that time, the industry was recovering between 10 and 20% of reservoir capacity, as compared to recent recovery rates of over 60% before wells become unproductive. The concept of control is that a faster rate of production more quickly dissipates the pressure in the reservoir, thereby reducing the total amount of oil which can be eventually recovered. Two measures used to conserve petroleum reservoirs are unitization and well spacing.

1. Unitization is the operation of a field as one unit in order to apply secondary recovery methods and maintain pressure, even through a number of different operators may be involved. The total production is allocated on an equitable basis among the operators.
2. Well spacing is the limiting and proper location of wells so as to achieve maximum production without dissipating a field due to overdrilling.

Methods of Recovering Additional Product

Productivity of oil and gas reservoirs is improved by a variety of recovery methods. One method is either to chemically or physically open passages in the strata to allow oil and gas to move more freely through reservoirs to the well. Water and gas are injected into reservoirs to maintain working pressure by natural displacement. Secondary recovery methods, including displacement by pressure, artificial lift and flooding, improve and restore reservoir pressure. Enhanced recovery is the use of various secondary recovery methods in multiple and different combinations. Enhanced recovery also includes more advanced methods of obtaining additional product from depleted reservoirs, such as thermal recovery, which uses heat instead of water or gas to force more crude oil out of reservoirs.

Acidizing

Acidizing is a method of increasing the output of a well by pumping acid directly into a producing reservoir to open flow channels through the reaction of chemicals and minerals. Hydrochloric (or regular) acid, was first used to dissolve limestone formations. It is still most commonly used; however, various chemicals are now added to the hydrochloric acid to control its reaction and to prevent corrosion and formation of emulsions.

Hydrofluoric acid, formic acid and acetic acid are also used, together with hydrochloric acid, depending on the type of rock or minerals in the reservoir. Hydrofluoric acid is always combined with one of the other three acids, and was originally used to dissolve sandstone. It is often called “mud acid”, as it is now used to clean perforations which have been plugged with drilling mud and to restore damaged permeability near the well hole. Formic and acetic acids are used in deep, ultra-hot limestone and dolomite reservoirs and as breakdown acids prior to perforation. Acetic acid is also added to wells as a neutralizing buffer agent to control the pH of well stimulation fluids. Almost all acids have additives, such as inhibitors to prevent reaction with the metal casings and surfactants to prevent formation of sludge and emulsions.

Fracturing

Fracturing describes the method used to increase the flow of oil or gas through a reservoir and into wells by force or pressure. Production may decrease because the reservoir formation is not permeable enough to allow the oil to flow freely toward the well. Fracturing forces open underground channels by pumping a fluid treated with special propping agents (including sand, metal, chemical pellets and shells) into the reservoir under high pressure to open fissures. Nitrogen may be added to the fluid to stimulate expansion. When the pressure is released, the fluid withdraws and the propping agents remain in place, holding the fissures open so that oil can flow more freely.

Massive fracturing (mass frac) involves pumping large amounts of fluid into wells to hydraulically create fissures which are thousands of feet in length. Massive fracturing is typically used to open gas wells where the reservoir formations are so dense that even gas cannot pass through them.

Pressure maintenance

Two common pressure maintenance techniques are the injection of water and gas (air, nitrogen, carbon dioxide and natural gas) into reservoirs where natural pressures are reduced or insufficient for production. Both methods require drilling auxiliary injection wells at designated locations to achieve the best results. The injection of water or gas to maintain the working pressure of the well is called natural displacement. The use of pressurized gas to increase the pressure in the reservoir is called artificial (gas) lift.

Water flooding

The most commonly used secondary enhanced recovery method is pumping water into an oil reservoir to push product toward producing wells. In five-spot water flooding, four injection wells are drilled to form a square with the producing well at the center. The injection is controlled to maintain an even advance of the water front through the reservoir toward the producing well. Some of the water used is salt water, obtained

from the crude oil. In low-tension water flooding, a surfactant is added to the water to assist the flow of oil through the reservoir by reducing its adhesion to rock.

Miscible flooding

Miscible fluid and miscible polymer flooding are enhanced recovery methods used to improve water injection by reducing the surface tension of crude oil. A fluid miscible (one that can be dissolved in the crude) is injected into a reservoir. This is followed by an injection of another fluid which pushes the crude and miscible fluid mixture toward the producing well. Miscible polymer flooding involves the use of a detergent to wash the crude oil from the strata. A gel or thickened water is injected behind the detergent to move the crude toward the producing well.

Fire flooding

Fire flooding, or in situ (in place) combustion, is an expensive thermal recovery method wherein large quantities of air or oxygen-containing gas is injected into the reservoir and a portion of the crude oil is ignited. The heat from the fire reduces the viscosity of the heavy crude oil so that it flows more easily. Hot gases, produced by the fire, increase the pressure in the reservoir and create a narrow burning front which pushes the thinner crude from the injection well to the producing well. The heavier crude remains in place, providing additional fuel as the flame front moves slowly forward. The burning process is closely monitored and controlled by regulating the injected air or gas.

Steam injection

Steam injection, or steam flooding, is a thermal recovery method which heats heavy crude oil and lowers its viscosity by injecting super-hot steam into the lowest stratum of relatively shallow reservoir. The steam is injected over a period of 10 to 14 days, and the well is shut for another week or so to allow the steam to thoroughly heat the reservoir. At the same time the increased heat expands reservoir gases, thereby increasing the pressure in the reservoir. The well is then reopened and the heated, less viscous crude flows up into the well. A newer method injects low-heat steam at lower

pressure into larger sections of two, three or more zones simultaneously, developing a "steam chest" which squeezes down the oil in each of the zones. This provides a greater flow of oil to the surface, while using less steam.

Environmental Protection:

The major sources of air, water and ground pollution in oil and natural gas production are from oil spills or gas leaks on land or sea, hydrogen sulphide present in oil and gas escaping into the atmosphere, hazardous chemicals present in drilling mud contaminating water or land and combustion products of oil well fires. The potential public health effects of inhalation of smoke particulates from large-scale oil field fires has been of great concern since the oil well fires that occurred in Kuwait during the Persian Gulf War in 1991.

Pollution controls typically include:

1. API separators and other waste and water treatment facilities
2. Spill control, including booms for spills on water
3. Spill containment, dikes and drainage to control oil spills and divert oily water to treatment facilities.

Gas dispersion modelling is conducted to ascertain the probable area which would be affected by a cloud of escaping toxic or flammable gas or vapour. Groundwater table studies are conducted to project the maximum extent of water pollution should oil contamination occur.

Workers should be trained and qualified to provide first aid response to mediate spills and leakage. Contractors who specialize in pollution remediation are usually engaged to manage large spill responses and remediation projects.

Questions:

1. What are the methods uses for locate the oil and natural gas deposits?
2. What is the composition of Crude Oil and Natural Gas? Explain.
3. How the oil and gas reservoirs is improved by a recovery method. Discuss.